

Limiti Notevoli

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\tan(x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \log_a e$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

ESERCIZIO n°1

$$\frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} \Rightarrow \frac{\sin^2 2 \cdot 0}{0^2} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$D[(\sin 2x)^2] = 2 \sin 4x$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin^2 2x}{x^2} \right)^2 \Rightarrow \left(\frac{\sin 2x}{x} \right)^2 = \frac{2 \sin 4x}{2x} = \frac{4x}{x} = 4$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} = 4$$

ESERCIZIO n°2

$$\frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{4x^2} = \frac{1 - \cos 3 \cdot 0}{4 \cdot 0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{4x^2} = \frac{D'(1 - \cos 3x)}{D'(4x^2)} = \frac{3 \sin 3x}{8x} \Rightarrow \frac{3 \cos 3x}{8}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{4x^2} = \frac{3 \cos 3x}{8} = \frac{3 \cos(3 \cdot 0)}{8} = \frac{3}{8}$$

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ESERCIZIO n° 3

$$\frac{\operatorname{Tg} x}{x} = 1$$

$$\frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$1 - \cos x = \frac{x^2}{2}$$

$$\lim_{x \rightarrow 0} \frac{3x \operatorname{Tg} x}{\cos 2x - 1} = \frac{3 \cdot 0 \cdot \operatorname{Tg} 0}{\cos 2 \cdot 0 - 1} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{3x \operatorname{Tg} x}{\cos 2x - 1} = 3x \cdot \operatorname{Tg} x \cdot \frac{x}{x} \cdot \frac{1}{\cos 2x - 1} \cdot \frac{-4x^2}{-4x^2}$$

$$\lim_{x \rightarrow 0} \frac{3x \operatorname{Tg} x}{\cos 2x - 1} = \frac{3x^3}{4x^2} \cdot 1 \cdot \frac{4x^2}{1 - \cos 2x} \stackrel{(\frac{1}{2})^{-1}}{\Rightarrow} - \frac{3 \cancel{6x^2}}{2 \cancel{4x^2}}$$

$$\lim_{x \rightarrow 0} \frac{3x \operatorname{Tg} x}{\cos 2x - 1} = - \frac{3 \cancel{6x^2}}{2 \cancel{4x^2}} = - \frac{3}{2}$$

ESERCIZIO n° 4

$$\frac{\sin x}{x} = 1$$

CRITERIO ASINTOTICHE

$$2 \sin x \approx 2x$$

$$3x \approx 3x$$

$$\sin 3x \approx 3x$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x + 3x}{\sin 3x - x} = \frac{2 \sin 0 + 3 \cdot 0}{\sin 3 \cdot 0 - 0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x + 3x}{\sin 3x - x} = \frac{2x + 3x}{3x - x} = \frac{5x}{2x} = \frac{5}{2}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x + 3x}{\sin 3x - x} \Rightarrow \frac{2 \cos x + 3}{3 \cos x - 1} = \frac{2 \cdot 1 + 3}{3 \cdot 1 - 1} = \frac{5}{2}$$

ESERCIZIO n° 5

$$\lim_{x \rightarrow 0} \frac{1 - \cos \sqrt{x} - \sin \sqrt{x}}{2x} = \frac{\sin \sqrt{x} - \cos \sqrt{x}}{2\sqrt{x}} = \frac{\sin \sqrt{x} - \cos \sqrt{x}}{4\sqrt{x} - \infty}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos \sqrt{x} - \sin \sqrt{x}}{2x} = -\infty$$

ESERCIZIO n°6

$$\lim_{x \rightarrow 0} \frac{x + x^2}{3x + \sin x} = \frac{0+0}{3 \cdot 0 + \sin 0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{x + x^2}{3x + \sin x} = \frac{x \cdot \overset{1}{(x+1)}}{3x + x} = \frac{1x}{4x} = \frac{1}{4}$$

ESERCIZIO n°7

$$\lim_{x \rightarrow 0} \frac{\sin 2x + x}{x + 3 \sin x} = \frac{\sin 2 \cdot 0 + 0}{0 + 3 \cdot \sin 0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

CRITERIO ASINTOTICHE

$$\sin x \approx x$$

$$\sin x \approx x$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x + x}{x + 3 \sin x} = \frac{2x + x}{x + 3x} = \frac{3x}{4x} = \frac{3}{4}$$

ESERCIZIO n°8

$$\frac{\sin x}{x} = 1$$

$$\frac{1 - \cos(2x)}{2} = \sin^2(x)$$

$$\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$$

$$\lim_{x \rightarrow 0^+} \ln(\tan x) - \ln(2x) = \infty - \infty$$

$$\lim_{x \rightarrow 0^+} \ln\left(\frac{\sin x}{\cos x}\right) - \ln(2x) = \ln\left(\frac{x \cdot \overset{1}{\frac{\sin x}{x}}}{\cos x}\right) - \ln(2x)$$

$$\lim_{x \rightarrow 0^+} \ln\left(\frac{x}{\cos x}\right) - \ln(2x) = \ln\left(\frac{x}{1 - 2 \sin^2\left(\frac{x}{2}\right)}\right) - \ln(2x)$$

$$\lim_{x \rightarrow 0^+} \left(-\frac{x}{\left(\frac{\sin\left(\frac{x}{2}\right)}{2}\right)^2 x^2} \right) - \ln(2x) = \ln\left(-\frac{2x}{x^2 - 2}\right) - \ln(2x)$$

$$\lim_{x \rightarrow 0^+} \ln\left(-\frac{1}{x^2 - 2}\right) \Rightarrow \ln\left(\frac{-1}{0^2 - 2}\right) = \ln\left(\frac{1}{2}\right) = -\ln(2)$$

ESERCIZIO n°9

$$\lim_{x \rightarrow 0} \frac{1 + \cos(x + \pi)}{\sin 2x} = \frac{1 + \cos(\pi)}{\sin 0} = \frac{1-1}{0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{1 + \cos(x + \pi)}{\sin 2x} = \frac{\sin x}{2 \cos 2x} = \frac{x}{4x} = \frac{0}{4 \cdot 0} = 0$$

ESERCIZIO n°10

$$\frac{1 - \cos(2\alpha)}{2} = \sin^2(\alpha)$$

$$\ln(\alpha) - \ln(\beta) = \ln\left(\frac{\alpha}{\beta}\right)$$

$$\lim_{x \rightarrow 0^+} \ln(\sin x) - \ln(\cos x - 1) = \infty - \infty$$

$$\lim_{x \rightarrow 0^+} 2 \ln(\sin x) - \ln(\cos x - 1) = 2 \ln(x) - \ln(-2 \sin^2(\frac{x}{2}))$$

$$\lim_{x \rightarrow 0^+} 2 \ln(x) - \ln(-\frac{x^2}{2}) \Rightarrow \ln(-2)$$

ESERCIZIO n°11

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{3x^2} = \frac{\sin 0 - \tan 0}{3 \cdot 0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin x - \frac{\sin x}{\cos x}}{x^2} = \frac{(\cos x - 1) \sin x}{x^2 \cos x}$$

$$\frac{1}{3} \lim_{x \rightarrow 0} \frac{(\cos x - 1) \sin x}{x^2} = \frac{(\cos x - 1) \frac{\sin x}{x}}{x} = \frac{\cos x - 1}{x} = \frac{-\sin x}{1}$$

$$\frac{1}{3} \lim_{x \rightarrow 0} -\sin x = 0$$

ESERCIZIO n°12

$$\lim_{x \rightarrow 0^+} \frac{e^{\sin x} - 1}{2x} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0^+} \frac{e^{\sin x} \cdot \cos x}{2} = \frac{1}{2}$$

ESERCIZIO n°13

$$\lim_{x \rightarrow 0^+} \frac{e^x - \cos x}{\sin 2x} = \frac{e^0 - \cos 0}{\sin 2 \cdot 0} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0^+} \frac{\sin x + e^x}{2 \cos 2x} = \frac{\sin 0 + e^0}{2 \cos 2 \cdot 0} = \frac{1}{2}$$

ESERCIZIO n°14

$$\alpha^{\beta} = e^{\beta \ln(\alpha)}$$

$$\frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \left(\frac{x}{\sin 2x} \right)^{x+2} = (0 \cdot \infty)^2$$

$$\lim_{x \rightarrow 0} \left(\frac{x}{\sin 2x} \right)^{x+2} = e^{(x+2) \ln \left(\frac{x}{\sin 2x} \right)} \Rightarrow (x+2) \ln \left(\frac{x}{\sin 2x} \right)$$

$$\lim_{x \rightarrow 0} -\ln 2 - 2 \ln 2 = -2 \ln 2$$

$$e^{(x+2) \ln \left(\frac{x}{\sin 2x} \right)} = \frac{1}{4}$$

ESERCIZIO n°15

$$\lim_{x \rightarrow +\infty} \left(\frac{x+1}{x-1} \right)^{3x} = \left(\frac{(x-1)+2}{x-1} \right)^{3x} = \left(\frac{x-1}{x-1} + \frac{2}{x-1} \right)^{3x} = \left(1 + \frac{2}{x-1} \right)^{3x} = \left(1 + \frac{2}{x-1} \cdot \frac{1}{\frac{1}{2}} \right)^{3x}$$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{2 \cdot \frac{1}{2}}{(x-1) \cdot \frac{1}{2}} \right)^{3x} = \left(1 + \frac{1}{\frac{1}{2}(x-1)} \right)^{\frac{1}{2}(x-1) \cdot 6+3} \quad t = \frac{1}{2}(x-1)$$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^{t \cdot 6+3} \Rightarrow \lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^{t \cdot 6} + \lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t} \right)^3 \Rightarrow e^6 \cdot 1 = e^6$$

ESERCIZIO n°16

$$\operatorname{Tg} x = \frac{\sin x}{\cos x}$$

$$\frac{1 - \cos 2\alpha}{2} = \sin^2 \alpha$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - \cos x}{2 \operatorname{Tg} x} = \frac{e^{\sin 0} - \cos 0}{2 \operatorname{Tg} \cdot 0} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - \cos x}{2 \operatorname{Tg} x} = \frac{e^{\sin x} - \cos x}{2 \left(\frac{\sin x}{\cos x} \right)} \Rightarrow \frac{1}{2} \lim_{x \rightarrow 0} \frac{(e^{\sin x} - \cos x) \cos x}{\sin x}$$

$$\frac{1}{2} \lim_{x \rightarrow 0} \frac{(e^{\sin x} - \cos x) \cos x}{\sin x} = -\frac{1}{2} \lim_{x \rightarrow 0} \frac{\cos^2 x - e^{\sin x} \cos x}{\sin x} = \frac{-\cos x (\cos x - e^{\sin x})}{\sin x}$$

$$-\frac{1}{2} \lim_{x \rightarrow 0} \frac{\cos x - e^{(\frac{\sin x}{x})}}{x - \frac{\sin x}{x}} \Rightarrow -\frac{1}{2} \lim_{x \rightarrow 0} \frac{\cos x - e^x}{x} \Rightarrow -\frac{1}{2} \lim_{x \rightarrow 0} \frac{e^x - 2 \sin^2 \left(\frac{x}{2} \right) + 1}{x}$$

$$\left(\sin \left(\frac{x}{2} \right) \right)^2$$

$$-\frac{1}{2} \lim_{x \rightarrow 0} \frac{-e^x - \frac{(\frac{e^x-1}{x})^2}{2} + 1}{x} = -\frac{1}{4} \lim_{x \rightarrow 0} \frac{2e^x + x^2 - 2}{2x} = -\frac{1}{4} \lim_{x \rightarrow 0} \frac{2-2(x \frac{e^x-1}{x} + 1)}{x} - x$$

$$-\frac{1}{4} \lim_{x \rightarrow 0} -x-2 \Rightarrow +\frac{1}{4} \lim_{x \rightarrow 0} 0+2 = \frac{1}{2} \quad +2(1+\frac{1}{x})$$

ESERCIZIO n°18

$$\lim_{x \rightarrow 0^+} \frac{a^x - 1}{x} = \ln a$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e^{\sqrt{4x}} - 1} = \frac{\sqrt{0}}{e^{\sqrt{4 \cdot 0}} - 1} = \frac{0}{0} \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e^{\sqrt{4x}} - 1} = \frac{\sqrt{x}}{e^{2\sqrt{x}} - 1} = \frac{\sqrt{x}}{(e^{\sqrt{x}} - 1)(e^{\sqrt{x}} + 1)} = \frac{\sqrt{x}}{e^{\sqrt{x}} - 1} \cdot \left(\frac{1}{2}\right) = \frac{1}{2} \lim_{x \rightarrow 0^+} \frac{1}{\frac{e^{\sqrt{x}} - 1}{\sqrt{x}}}$$

$$\frac{1}{2} \lim_{x \rightarrow 0^+} 1 = \frac{1}{2}$$

ESERCIZIO n°19

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x+1}\right)^{3x+2} = \left(\frac{\infty}{\infty+1}\right)^{3 \cdot \infty + 2} = 0 \cdot \infty$$

$$\frac{x}{x+1} = 1 - \frac{1}{x+1}$$

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x+1}\right)^{3x+2} = \left(1 - \frac{1}{x+1}\right)^{3x+2}$$

$$3x+2 = (-x-1)(-3)-1$$

$$\lim_{x \rightarrow \infty} (1)^{(-x-1)(-3)-1} = \frac{1}{e^3} \Rightarrow e^{-3}$$

ESERCIZIO n°20

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0^+} (1+3x)^{\frac{1}{x}} = 1^\infty$$

$$\frac{1}{x} = \frac{1}{\frac{1}{3x}} = 3$$

$$\lim_{x \rightarrow 0^+} (1+3x)^{\frac{1}{x}} = (1+3x)^{\frac{1}{\frac{1}{3x}} \cdot 3} = \lim_{x \rightarrow 0} e^3$$

ESERCIZIO n°21

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{e^{2+x} - e^{2x}} = \frac{1 - \cancel{\cos 0}}{e^{2+0} - e^{2 \cdot 0}} = 0$$

ESERCIZIO n°22

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+5} - \sqrt{x+2}}{2} = \infty - \infty$$